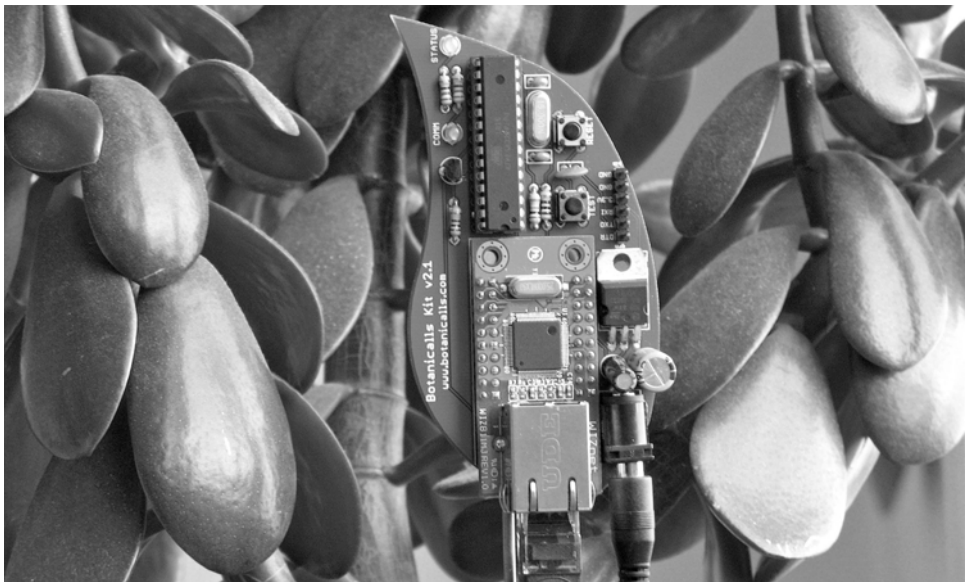


Figure 1-6. *Botanicalls Arduino-based tweeting plant water meter, courtesy Botanicalls*

These projects and many others have all benefited from the rapid development and reliable longevity brought to you by Arduinoland and the hard work put in by the Arduino team and community at large to create an open architecture that enables continued and accelerated growth and creativity.

Arduino is C… Mostly

In addition to the various aspects of the diverse Arduino ecosystem, we have the programming language of the Arduino platform, which is the central focus of this book. The core language used in the Arduino development environment is the C computer programming language first developed at the research institute of Bell Laboratories in the early 1970s for use with the UNIX operating system. C uses a procedural language syntax that needs to be processed by a compiler to map human-readable code to machine instructions. The long-standing popularity of C lends the Arduino some of its heritage, but the code that we are writing in this book is only mostly C.

Because there are aspects of the C language that look like it was written by dyslexic aliens, and with the language sometimes accused of being overly cryptic and difficult for beginners to pick up, the Arduino team has developed the standard Arduino library that provides a simple and targeted set of functions that make programming the Arduino interface board about as easy as it can get. Now, these libraries are themselves actually C++, itself a subset of the original C language, but we really don't need to go there.

What's important is that most of the code that we will write for the Arduino, including its syntax, structure, operators, control statements, and functions, remain fundamentally and functionally the same as C. What will be unique to the Arduino, however, are all sorts of functions that you will come to know and love, including `pinMode()`, `digitalWrite()`, and `delay()` that are specific to the standard Arduino library. For the purposes of this book, this basic framework of C combined with the additional Arduino library that is automatically a part of every sketch that we write, is what we will refer to as